

Machine Learning from Scratch – WiDS Project report

This is a summary of the WiDS project that I worked on over the last few months, titled “Machine Learning from Scratch.” I was inclined to undertake this project primarily because I have always been very curious about machine learning and artificial intelligence, and I wanted to understand how AI truly works beneath the surface. While AI is often spoken about in very broad and sometimes abstract terms, I wanted to go deeper into the foundations and understand the mathematical and conceptual ideas that make these systems function.

One of the most positive aspects of this project was the structured manner in which it began. Instead of immediately diving into complex models or coding implementations, we started with an introduction to what many commonly used jargon terms actually mean. In today’s environment, words like AI, machine learning, and deep learning are used very freely, often without a clear or consistent explanation. Reading about these concepts from the ground up helped demystify them. Through these introductory readings, I gained clarity on how these terms differ from one another and how they are related, which made the later stages of the project much easier to follow.

Initially, we were given a set of articles to read, which allowed us to explore the precise definitions and boundaries of these fields. This was particularly enlightening because it forced me to slow down and think about what each term truly encompasses, rather than relying on assumptions or popular explanations. Over time, this helped me develop a more intuitive understanding of what artificial intelligence includes, where machine learning fits within it, and how deep learning emerges as a specialized subset. This conceptual clarity was important because it laid a strong foundation for everything that followed.

Following the readings, we moved on to a structured video series that further expanded on these ideas. These videos were helpful in visually and practically explaining how the different components of AI are classified. They discussed the scope of artificial intelligence, the terminologies used within the field, what machine learning is capable of doing, and equally importantly, what it cannot do. This distinction was especially useful because it prevented unrealistic expectations about AI systems. The series also gave us insight into where current technologies such as GPT-based systems stand today and how they fit into the broader evolution of AI. Understanding the limitations and current capabilities of these systems gave me a more grounded perspective on modern AI developments.

After building this conceptual understanding, the project transitioned into revisiting foundational mathematical tools, particularly linear algebra and probability. This part of the project felt both challenging and enjoyable. It served as a revision of concepts I had previously studied during my JEE preparation, but now with a completely different motivation and context. Revisiting vectors, matrices, probability distributions, and expectations was much more engaging because I could clearly see how these ideas are directly applied in machine learning models. Instead of being abstract mathematical exercises, they became tools for understanding and building algorithms.

As we progressed further, the assignments themselves guided us into exploring linear regression in detail. This was particularly interesting for me because during my first semester studying IEOR, I had already encountered sections of statistics that dealt with regression analysis. However, this project allowed me to revisit those ideas with a stronger emphasis on interpretation and application. I found it very rewarding to extrapolate what I had learned previously and apply it to real machine learning contexts. Understanding how linear regression works from both a

statistical and algorithmic perspective gave me deeper insight into how learning from data actually occurs.

A significant part of this learning process involved focusing on the mathematical structure of regression models. Instead of treating regression as a black-box method, the project emphasized understanding loss functions, optimization, and how parameters are updated. This approach helped me appreciate why linear regression is often considered the foundation of many machine learning algorithms. It also highlighted how small mathematical ideas, when scaled appropriately, can lead to powerful predictive models.

I also greatly enjoyed studying topics through the book “An Introduction to Statistical Learning (ISLV).” The book provided a clear and intuitive explanation of many machine learning concepts without overwhelming the reader with excessive mathematical rigor. It bridged the gap between theory and practice very effectively. Reading ISLV alongside the assignments helped reinforce what we were learning and gave me additional examples and perspectives on how these models are used in real-world scenarios.

As the project progressed and we moved toward understanding how machine learning algorithms work in general, one of the biggest clarities for me was truly understanding what the term regression refers to. Before this project, I had encountered the term many times, but I had never fully grasped its meaning. Through the assignments and readings, I came to understand regression as a method of modeling relationships between variables, rather than just a computational technique. This conceptual shift was important because it changed how I interpret results and evaluate models.

Another realization that stood out to me was the distinction between quantitative and qualitative approaches to handling data. There is a meaningful difference between what can be represented numerically and what must be represented probabilistically or categorically. This helped clarify why certain problems are better suited to regression, while others require classification methods. For example, a medical diagnosis often involves a yes-or-no decision, which naturally falls under classification, whereas something like predicting the salaries of baseball hitters depends on continuous numerical variables and is more suitable for regression. This distinction helped me better understand how datasets are structured and how models should be chosen accordingly.

One of the most interesting topics covered in the project was ensemble methods. Understanding how different models can be combined to improve performance was both conceptually rich and practically useful. Learning about methods such as bagging, boosting, and random forests showed me how training dynamics change when multiple learners interact. While much of this topic was theoretical, it provided a lot of conceptual clarity about variance reduction, bias, and overfitting. I also explored additional simulations outside the coursework, which helped reinforce these ideas visually and intuitively. These explorations made ensemble methods one of the areas I am most interested in studying further on my own.

Decision trees were another component of the project that I found particularly engaging. Their structure is intuitive and interpretable, which makes them an excellent tool for understanding how decisions are made within a model. Alongside decision trees, we studied the confusion matrix, which was especially interesting because of its applications in diagnostic settings. I had previously encountered the confusion matrix in my first-semester biology course, but studying it now from a statistical and analytical perspective gave it a completely new meaning. Calculating metrics such as accuracy, precision, recall, and understanding why certain

metrics matter more than others in specific contexts felt like a meaningful application of interdisciplinary knowledge.

In addition to these classical machine learning topics, the assignments also introduced concepts related to neural networks, which added another important dimension to the project. One of the key ideas explored was the Rectified Linear Unit (ReLU) activation function. Understanding ReLU helped clarify how non-linearity is introduced into neural networks and why it is so widely used in practice. Compared to older activation functions, ReLU's simplicity and effectiveness in avoiding vanishing gradients made it easier to see how deep networks can be trained efficiently.

I also referred to a 3B1B playlist of neural networks to understand the content in more depth, which was very enlightening and fun, as the visualizations helped me truly understand everything better.

We also explored classification in neural networks, particularly how outputs are interpreted as probabilities rather than direct numerical predictions. This tied back nicely to earlier discussions about qualitative versus quantitative outcomes. Seeing how neural networks handle classification problems using concepts such as softmax outputs and decision boundaries helped me connect earlier regression-based ideas with more complex learning architectures. It reinforced the idea that many advanced models are built upon relatively simple principles, combined in structured ways.

Another important aspect of studying neural networks was understanding how learning actually occurs through loss functions and gradient-based optimization. Tracing how errors propagate backward through a network gave me a clearer picture of how models adjust themselves during training. This understanding complemented earlier work on linear regression and

optimization, showing how similar ideas are extended to more complex systems.

Overall, working through neural-network-based questions helped solidify my understanding of how modern machine learning systems operate. Rather than viewing neural networks as opaque or purely computational tools, I began to see them as structured mathematical models that follow logical principles. This perspective made the subject far less intimidating and much more approachable.

In conclusion, this project has been a highly enriching learning experience. Starting from fundamental definitions and gradually progressing to advanced models allowed me to build confidence and understanding step by step. I particularly appreciated how the project emphasized intuition, mathematical reasoning, and real-world relevance over rote implementation. By the end of the project, I felt that I not only learned how machine learning models work, but also why they work the way they do. This whole project has motivated me to continue exploring the field further and apply these concepts to more complex and practical problems in the future, and i really really look forward to doing it!

Thanks & Regards,

Saanvi Walvekar